

Curriculum Vitae

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Profile

- Engineer with a solid background in software, data engineering, and machine learning
- Value-driven, versatile, and resilient professional
- Proven track record of delivering scalable solutions using agile methodologies
- Focus on automation, maintainability, best practices, and design patterns
- Thrives in dynamic environments with cross-functional teams
- Committed to applying and developing knowledge to solve real-world challenges in software, data, and machine learning domains

Main Skills

- **Programming Languages & Frameworks:** Python, PySpark, Java, JavaScript/Node.js/React/Angular, HTML/CSS/XML, R
- **Data & Analytics:** Databricks, Apache Airflow, Statistical Process Control, Machine Learning, Deep Learning
- **Databases:** SQL/RDBMS, Oracle, MongoDB, Cloudant, MS SQL Server, IBM DB2, Cassandra, PostgreSQL, DynamoDB
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Helm, Azure DevOps, Travis CI, Terraform, AWS-CDK
- **Monitoring & Tools:** Datadog, Application Insights, OpenTelemetry
- **Other:** Linux/Bash Scripting, UML, Spring (Core/Boot/MVC/Data-JPA), JAX-WS & JAX-RS, JPA/Hibernate

Professional Experience

Data Engineer

Novo Nordisk, Søborg, Denmark

Nov 2023 – Present

- Designed, developed, and maintained GxP critical and non-critical data analytics solutions in the Safety Surveillance Data Science team
- Implemented data product solutions including project templates, persistence layer, data pipeline abstraction constructs, observability, data quality and data contracts

Technologies & Methodologies: AWS-CDK, AWS, Azure, Python, PySpark, Pandas, PyArrow, Databricks, Databricks Asset Bundles, Azure DevOps, Terraform, R, Bash.

Machine Learning Engineer

Pandora A/S, Copenhagen, Denmark

Oct 2020 – Nov 2023

- Designed, interfaced, and integrated data products in the Data Science & Machine Learning team
- Planned and provided technical steering as product owner
- Designed and implemented MLOps blueprints to support data science adoption and scaling in the organization

Technologies & Methodologies: Apache Airflow, Python, Apache PySpark, Databricks, Data Lake, Azure, Azure DevOps, Azure Functions, K8s, Terraform, Bash, SAFe, Java, Apache Beam.

Software Engineer

Digital, A.P. Moller - Maersk A/S, Copenhagen, Denmark
Feb 2019 – Oct 2020

- Implemented and maintained microservice APIs in the Future Pricing and Optimization team

Technologies & Methodologies: Java 8, Spring Boot, Hibernate, Swagger, Microsoft SQL Server, Azure, Docker, Scrum, TDD, Terraform, Kubernetes, Helm, Azure DevOps, Jenkins.

Software Engineer

8 West Consulting, Cork, Ireland
Nov 2017 – Feb 2019

- Implemented a data layer microservice API for Data Provider Accuracy

Technologies & Methodologies: Java 8, Spring Boot, Hibernate, Swagger, RAML, JPA, Oracle, Docker.

Software Engineer

IBM, Cork, Ireland
Jan 2014 – Nov 2017

- Built microservices and web UI for a cognitive application (Watson NLP, machine learning orchestration)
- Developed and maintained analytics components across Cloud and Smarter Infrastructure monitoring products
- Conducted proof-of-concept experiments; analyzed algorithm performance on real-time data sets

Technologies & Methodologies: Java, Dropwizard, Spring, Python, Node.js, Angular, IBM Watson, Cloudant, MongoDB, Redis, Docker, IBM Bluemix, Cloud Foundry, Statistical Process Control, R.

Software Engineer

DeCare Systems, Cork, Ireland
Jul 2013 – Jan 2014

- Maintained and implemented bug fixes in a health care management web application
- Configured and programmed an ANSI 834 Health Care Enrollment (HIPAA compliant) middleware data interface

Technologies & Methodologies: Java, Struts, JavaScript, JSP, Hibernate, Oracle, Design Patterns; IBM WebSphere Transformation Extender (WTX), Oracle.

Software Engineer

Wincor Nixdorf, Lisbon, Portugal
Aug 2008 – Jul 2013

- Developed retail web application; designed and built a fiscal library component ensuring legal compliance and certification support
- Handled international roll-out, support, and maintenance; developed a mobile & automated checkout solution for a Portuguese retailer

Technologies & Methodologies: Java, Spring MVC, JPA/Hibernate, C, C#, JavaScript, PostgreSQL, MS SQL Server, Derby, Linux, Bash, UML, TDD, JUnit.

Early Career

Various companies, Portugal & Ireland
Jan 2004 – Aug 2008

- EID S.A. (2007–2008): Implemented GUI Management Console for routing tables in an MMHS infrastructure (Java, Swing, UML, TDD)

- AXA Insurances (2006–2007): Developed call-centre script-guide software and maintained organizational process solutions (VB6, C#, SQL, Oracle)
- Care4IT (2004–2006): Analyzed, designed, and implemented healthcare management solutions (Java, Swing, SQL)
- Sival, Quality Engineer internship (2004): Implemented Statistical Process Control on production processes (SPC, control charts, multivariate analysis)

Educational Qualifications

Higher Diploma in Data Science and Analytics

Cork Institute of Technology (Munster Technological University) Feb 2017 – Dec 2018

Subjects: Statistics & Mathematics, Computer Science and Data Science, Time Series, Multivariate Analysis, Data Mining & Knowledge Discovery, Data Visualization, Machine Learning, Deep Learning.

Post-Graduate Studies in Lean Management

NOVA School of Science and Technology, NOVA University Lisbon

Oct 2008 – Jul 2009

Subjects: Lean Thinking, Value Stream Mapping, TPM, SMED, JIT Systems, Six-Sigma, Kaizen Events, Lean Supply Chain Management.

Industrial Engineering University Degree

NOVA School of Science and Technology, NOVA University Lisbon

Sep 1996 – Dec 2004

Subjects: Production Management, Industrial Logistics, Quality Engineering, Ergonomics, IT, Industrial Automation, Computer Aided Production and Manufacturing Technology.

References

Excellent references available on request.